# MSG No. 138.530 $P_I4_2/ncm$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `4'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,6,7,8,18,19,20,21]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[2,3,4,5,17,22,23,24]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,14,15,16,26,27,28,29]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[10,11,12,13,25,30,31,32]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[1,6,15,16,20,21,26,27]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[2,3,12,13,23,24,25,30]` |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[4,5,10,11,17,22,31,32]` |
| 4 | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[7,8,9,14,18,19,28,29]` |

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1,6,10,11,18,19,25,30]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,9,14,17,22,26,27]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[4,5,15,16,23,24,28,29]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[7,8,12,13,20,21,31,32]` |

Table 4: Wyckoff site: `4d`, site symmetry: `m'.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[1,6,15,16,23,24,25,30]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,12,13,20,21,26,27]` |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[4,5,10,11,18,19,28,29]` |
| 4 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[7,8,9,14,17,22,31,32]` |

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | `[1,7,9,15]` |
| 2 | $\left[0, 0, \frac{1}{2}\right]$ | `[2,5,10,13]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[3,4,11,12]` |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | `[6,8,14,16]` |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[17,23,25,31]` |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[18,21,26,29]` |
| 7 | $[0, 0, 0]$ | `[19,20,27,28]` |

| | | |
|---|---|---|
| 8 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[22,24,30,32]` |

Table 6: Wyckoff site: `8f`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$ | `[1,6,18,19]` |
| 2 | $\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[2,3,17,22]` |
| 3 | $\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$ | `[4,5,23,24]` |
| 4 | $\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$ | `[7,8,20,21]` |
| 5 | $\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$ | `[9,14,26,27]` |
| 6 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$ | `[10,11,25,30]` |
| 7 | $\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$ | `[12,13,31,32]` |
| 8 | $\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$ | `[15,16,28,29]` |

Table 7: Wyckoff site: `8g`, site symmetry: `2.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$ | `[1,6,15,16]` |
| 2 | $\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[2,3,12,13]` |
| 3 | $\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$ | `[4,5,10,11]` |
| 4 | $\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$ | `[7,8,9,14]` |
| 5 | $\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$ | `[17,22,31,32]` |
| 6 | $\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$ | `[18,19,28,29]` |
| 7 | $\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$ | `[20,21,26,27]` |
| 8 | $\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$ | `[23,24,25,30]` |

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x,\ \frac{3}{4}\right]$ | `[1,16,23,30]` |
| 2 | $\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$ | `[2,12,21,27]` |
| 3 | $\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | `[3,13,20,26]` |
| 4 | $\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$ | `[4,10,19,29]` |
| 5 | $\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | `[5,11,18,28]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | `[6,15,24,25]` |
| 7 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | `[7,14,17,32]` |
| 8 | $\left[-x,\ -x,\ \frac{1}{4}\right]$ | `[8,9,22,31]` |

Table 9: Wyckoff site: `16i`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x+\frac{1}{2},\ 0\right]$ | `[1,7]` |
| 2 | $\left[-x,\ x,\ \frac{1}{2}\right]$ | `[2,5]` |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[3,4]` |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2}-x,\,-x,\,0\right]$ | [6,8] |
| 5 | $\left[-x,\,\frac{1}{2}-x,\,0\right]$ | [9,15] |
| 6 | $\left[x,\,-x,\,\frac{1}{2}\right]$ | [10,13] |
| 7 | $\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,\frac{1}{2}\right]$ | [11,12] |
| 8 | $\left[x+\frac{1}{2},\,x,\,0\right]$ | [14,16] |
| 9 | $\left[x+\frac{1}{2},\,x,\,\frac{1}{2}\right]$ | [17,23] |
| 10 | $\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,0\right]$ | [18,21] |
| 11 | $[x,\,-x,\,0]$ | [19,20] |
| 12 | $\left[-x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$ | [22,24] |
| 13 | $\left[\frac{1}{2}-x,\,-x,\,\frac{1}{2}\right]$ | [25,31] |
| 14 | $\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,0\right]$ | [26,29] |
| 15 | $[-x,\,x,\,0]$ | [27,28] |
| 16 | $\left[x,\,x+\frac{1}{2},\,\frac{1}{2}\right]$ | [30,32] |

Table 10: Wyckoff site: 16j, site symmetry: .2'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\,\frac{1}{4},\,0\right]$ | [1,20] |
| 2 | $\left[\frac{1}{4},\,x,\,\frac{1}{2}\right]$ | [2,23] |
| 3 | $\left[\frac{1}{4},\,\frac{1}{2}-x,\,\frac{1}{2}\right]$ | [3,24] |
| 4 | $\left[x+\frac{1}{2},\,\frac{3}{4},\,\frac{1}{2}\right]$ | [4,17] |
| 5 | $\left[-x,\,\frac{3}{4},\,\frac{1}{2}\right]$ | [5,22] |
| 6 | $\left[\frac{1}{2}-x,\,\frac{1}{4},\,0\right]$ | [6,21] |
| 7 | $\left[\frac{3}{4},\,x+\frac{1}{2},\,0\right]$ | [7,18] |
| 8 | $\left[\frac{3}{4},\,-x,\,0\right]$ | [8,19] |
| 9 | $\left[-x,\,\frac{3}{4},\,0\right]$ | [9,28] |
| 10 | $\left[\frac{3}{4},\,-x,\,\frac{1}{2}\right]$ | [10,31] |
| 11 | $\left[\frac{3}{4},\,x+\frac{1}{2},\,\frac{1}{2}\right]$ | [11,32] |
| 12 | $\left[\frac{1}{2}-x,\,\frac{1}{4},\,\frac{1}{2}\right]$ | [12,25] |
| 13 | $\left[x,\,\frac{1}{4},\,\frac{1}{2}\right]$ | [13,30] |
| 14 | $\left[x+\frac{1}{2},\,\frac{3}{4},\,0\right]$ | [14,29] |
| 15 | $\left[\frac{1}{4},\,\frac{1}{2}-x,\,0\right]$ | [15,26] |
| 16 | $\left[\frac{1}{4},\,x,\,0\right]$ | [16,27] |

Table 11: Wyckoff site: 16k, site symmetry: m'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\,y,\,\frac{3}{4}\right]$ | [1,30] |
| 2 | $\left[\frac{1}{2}-y,\,x,\,\frac{1}{4}\right]$ | [2,27] |
| 3 | $\left[y,\,\frac{1}{2}-x,\,\frac{1}{4}\right]$ | [3,26] |
| 4 | $\left[x+\frac{1}{2},\,-y,\,\frac{3}{4}\right]$ | [4,29] |
| 5 | $\left[-x,\,y+\frac{1}{2},\,\frac{3}{4}\right]$ | [5,28] |
| 6 | $\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,\frac{3}{4}\right]$ | [6,25] |
| 7 | $\left[y+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{4}\right]$ | [7,32] |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 8 | $\left[-y,\, -x,\, \frac{1}{4}\right]$ | [8,31] |
| 9 | $\left[-x,\, -y,\, \frac{1}{4}\right]$ | [9,22] |
| 10 | $\left[y+\frac{1}{2},\, -x,\, \frac{3}{4}\right]$ | [10,19] |
| 11 | $\left[-y,\, x+\frac{1}{2},\, \frac{3}{4}\right]$ | [11,18] |
| 12 | $\left[\frac{1}{2}-x,\, y,\, \frac{1}{4}\right]$ | [12,21] |
| 13 | $\left[x,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$ | [13,20] |
| 14 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{4}\right]$ | [14,17] |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$ | [15,24] |
| 16 | $\left[y,\, x,\, \frac{3}{4}\right]$ | [16,23] |

Table 12: Wyckoff site: 16l, site symmetry: ..m

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, x,\, z]$ | [1,16] |
| 2 | $\left[\frac{1}{2}-x,\, x,\, z+\frac{1}{2}\right]$ | [2,12] |
| 3 | $\left[x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [3,13] |
| 4 | $\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$ | [4,10] |
| 5 | $\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [5,11] |
| 6 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z\right]$ | [6,15] |
| 7 | $\left[x+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$ | [7,14] |
| 8 | $[-x,\, -x,\, -z]$ | [8,9] |
| 9 | $\left[x+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [17,32] |
| 10 | $\left[-x,\, x+\frac{1}{2},\, z\right]$ | [18,28] |
| 11 | $\left[x+\frac{1}{2},\, -x,\, z\right]$ | [19,29] |
| 12 | $\left[x,\, \frac{1}{2}-x,\, -z\right]$ | [20,26] |
| 13 | $\left[\frac{1}{2}-x,\, x,\, -z\right]$ | [21,27] |
| 14 | $\left[-x,\, -x,\, z+\frac{1}{2}\right]$ | [22,31] |
| 15 | $\left[x,\, x,\, \frac{1}{2}-z\right]$ | [23,30] |
| 16 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [24,25] |

Table 13: Wyckoff site: 32m, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$ | [4] |
| 5 | $\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [5] |
| 6 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$ | [6] |
| 7 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$ | [7] |
| 8 | $[-y,\, -x,\, -z]$ | [8] |
| 9 | $[-x,\, -y,\, -z]$ | [9] |
| 10 | $\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [11] |

*continued ...*

Table 13

| No. | position | mapping |
|---|---|---|
| 12 | $\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [13] |
| 14 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$ | [14] |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$ | [15] |
| 16 | $[y,\, x,\, z]$ | [16] |
| 17 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [17] |
| 18 | $\left[-y,\, x+\frac{1}{2},\, z\right]$ | [18] |
| 19 | $\left[y+\frac{1}{2},\, -x,\, z\right]$ | [19] |
| 20 | $\left[x,\, \frac{1}{2}-y,\, -z\right]$ | [20] |
| 21 | $\left[\frac{1}{2}-x,\, y,\, -z\right]$ | [21] |
| 22 | $\left[-x,\, -y,\, z+\frac{1}{2}\right]$ | [22] |
| 23 | $\left[y,\, x,\, \frac{1}{2}-z\right]$ | [23] |
| 24 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [24] |
| 25 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [25] |
| 26 | $\left[y,\, \frac{1}{2}-x,\, -z\right]$ | [26] |
| 27 | $\left[\frac{1}{2}-y,\, x,\, -z\right]$ | [27] |
| 28 | $\left[-x,\, y+\frac{1}{2},\, z\right]$ | [28] |
| 29 | $\left[x+\frac{1}{2},\, -y,\, z\right]$ | [29] |
| 30 | $\left[x,\, y,\, \frac{1}{2}-z\right]$ | [30] |
| 31 | $\left[-y,\, -x,\, z+\frac{1}{2}\right]$ | [31] |
| 32 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [32] |